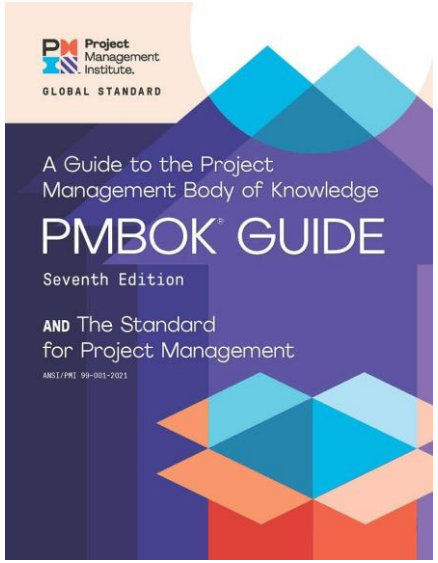
IT3120- Lecture 11

Project Management – Procurement, Resource and Schedule Management

Activities, Tools, Techniques, Inputs and Outputs related to Procurement Management, Resource Management and Schedule Management



Lecture Objectives

- Explain procurement, resource, and schedule management concepts.
- Identify key activities and processes involved.
- Apply PMBOK practices to software projects.
- Analyze practical Sri Lankan project scenarios.
- Understand how ITTOs support project success.

What is Procurement Management?

- Procurement Management in project management refers to the processes and activities involved in acquiring products, services, or resources from external sources outside the project team.

- Procurement management ensures that organizations obtain the necessary goods and services required to successfully complete a project while managing cost, quality, risk, and supplier relationships.



Main Activities in Procurement Management

- Following are the major procurement management activities:


Planning Procurement


Conducting Procurements


Controlling Procurements


Closing Procurement

Main Activities in Procurement Management



Planning Procurement

- Planning Procurement is the process of determining what products, services, or resources need to be obtained from external sources and deciding how those procurements will be managed throughout the project lifecycle.

- Procurement planning helps ensure that the project acquires the right goods and services at the right time, cost, and quality while supporting overall project value delivery.



Planning Procurement (Contd.)

Key Activities in Procurement Planning

- **Identifying Procurement Requirements**
  - Determine which goods or services are required externally.
  - Example: purchasing servers, cloud hosting, or hiring external developers.
- **Make-or-Buy Analysis**
  - Decide whether to develop internally or purchase from vendors.
- **Defining Procurement Strategy**
  - Select contract types and procurement approaches.
- **Preparing Procurement Documents**
  - Create Request for Proposal (RFP), Request for Quotation (RFQ), or Statements of Work (SOW).
- **Defining Vendor Selection Criteria**
  - Establish evaluation factors such as cost, experience, quality, and technical capability.

Main Activities in Procurement Management



Conducting Procurements

- Conducting Procurements is the process of obtaining seller responses, selecting suppliers or vendors, and awarding contracts for products or services required for a project.

- This process focuses on establishing effective relationships with external suppliers to ensure successful project delivery and value creation.

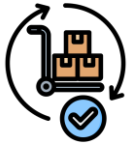


Conducting Procurements (Contd.)

Key Activities in Conducting Procurements

1. **Issuing Procurement Documents**
    - Sending Requests for Proposals (RFPs), Requests for Quotations (RFQs), or Invitations to Bid to potential vendors.
2. **Receiving Seller Responses**
    - Collecting proposals, quotations, and technical submissions from suppliers.
3. **Evaluating Proposals**
    - Assessing vendors based on cost, quality, experience, technical capability, and delivery schedules.
4. **Negotiating Contracts**
    - Discussing pricing, timelines, service levels, and legal terms before final agreement.
5. **Selecting Sellers**
    - Choosing the most suitable vendor and awarding the contract.

Main Activities in Procurement Management



Controlling Procurements

- Controlling Procurements is the process of monitoring procurement relationships, managing contracts, evaluating seller performance, and ensuring that procured products or services meet project requirements.

- This process helps ensure that suppliers fulfill contractual obligations while supporting successful project delivery and value realization.



Controlling Procurements (Contd.)

Key Activities in Controlling Procurements

- **Monitoring Seller Performance**
  - Track whether vendors are meeting schedule, cost, quality, and service requirements.
- **Managing Contracts**
  - Ensure both buyer and seller comply with agreed contractual terms.
- **Reviewing Deliverables**
  - Validate that procured products or services meet project specifications.
- **Handling Procurement Changes**
  - Process contract modifications or procurement-related adjustments when necessary.
- **Managing Payments and Claims**
  - Verify invoices, approve payments, and address disputes or claims.

Main Activities in Procurement Management



Closing Procurement

- Closing Procurements is the process of completing and finalizing all procurement activities and contractual obligations associated with a project.

- This process ensures that procurement agreements are formally completed, accepted, and documented while capturing lessons learned for future projects.



Closing Procurement (Contd.)

- **Key Activities in Closing Procurements**
- **Verifying Deliverables**
  - Confirm that all products or services have been delivered according to contract requirements.
- **Conducting Procurement Audits**
  - Review procurement performance and identify strengths and weaknesses.
- **Finalizing Payments**
  - Ensure all invoices and financial obligations are settled.
- **Archiving Procurement Records**
  - Store contracts, reports, correspondence, and procurement documentation for future reference.
- **Documenting Lessons Learned**
  - Capture procurement successes, issues, and recommendations for future projects.



In Class Activity 1

Multiple Choice Question

Which procurement process focuses on selecting vendors and awarding contracts?

A. Plan Procurement Management

B. Control Procurements

C. Conduct Procurements

D. Close Procurements



Multiple Choice Question

Which of the following is the BEST example of a procurement risk in an IT project?

A. Lack of software testing

B. Vendor dependency and delayed deliveries

C. Poor team communication

D. Incorrect activity sequencing



Multiple Choice Question

During procurement control, a project manager mainly focuses on:

A. Creating the project charter

B. Monitoring vendor performance and contract compliance

C. Estimating project durations

D. Recruiting internal project staff



What is Resource Management?

- Resource Management in project management refers to the processes involved in identifying, acquiring, developing, managing, and controlling the resources needed to successfully complete a project.

- Effective resource management ensures that the right resources are available at the right time and are utilized efficiently to achieve project objectives and deliver value.



Main Activities in Resource Management



Planning Resource Management



Estimating Activity Resources



Acquiring Resources



Developing the Team



Managing the Team



Controlling Resources

Main Activities in Resource Management



Planning Resource Management

- Planning Resource Management is the process of defining how project resources will be estimated, acquired, managed, utilized, and controlled throughout the project lifecycle.

- This process helps ensure that the project has the right people and physical resources available at the right time to achieve project objectives effectively and efficiently.



Planning Resource Management (Contd.)

Key Activities in Planning Resource Management

- **Identifying Resource Needs**
  - Determine the human resources, equipment, facilities, and materials required for project activities.
- **Defining Roles and Responsibilities**
  - Clarify team member duties, reporting relationships, and authority levels.
- **Developing Team Management Strategies**
  - Plan training, performance evaluations, motivation, and conflict management approaches.
- **Creating Resource Management Procedures**
  - Define how resources will be allocated, monitored, and controlled during the project.
- **Planning Communication and Collaboration**
  - Establish communication methods and team interaction guidelines.

Main Activities in Resource Management

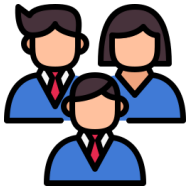


Estimating Activity Resources

- Estimating Activity Resources is the process of determining the type and quantity of resources required to complete project activities successfully.

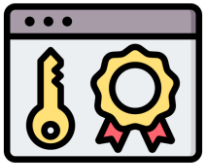
- This process helps project managers ensure that adequate human resources, equipment, materials, and technology are available to perform project work efficiently and achieve project objectives.

Types of Resources Estimated



Human Resources
(Developers, Testers, Analysts)





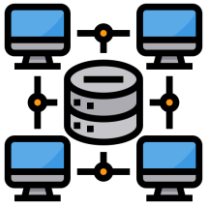
Software Tools and Licenses



Facilities and Infrastructure



Materials and Equipment



Physical Resources
(Servers, Laptops, Networking Devices)



Estimating Activity Resources(Contd.)

Key Activities in Estimating Activity Resources

- **Reviewing Project Activities**
  - Analyze project tasks and work packages.
- **Identifying Required Resources**
  - Determine what resources are needed for each activity.
- **Estimating Resource Quantities**
  - Estimate the number of people, equipment, or materials required.
- **Assessing Resource Availability**
  - Check resource availability and organizational constraints.
- **Documenting Resource Requirements**
  - Record resource estimates for scheduling and budgeting purposes.

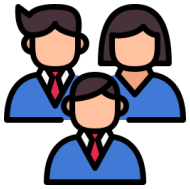
Main Activities in Resource Management



Acquiring Resources

- Acquiring Resources is the process of obtaining the team members, equipment, materials, facilities, and other resources necessary to complete project work successfully.
- This process ensures that the project has the required resources available at the right time to support efficient project execution and value delivery.
- The main purpose of acquiring resources is to:
  - Secure the required human and physical resources for the project
  - Ensure resources are available according to the project schedule
  - Improve project productivity and performance
  - Support timely project delivery
  - Address resource shortages and constraints

Types of Resources Acquired



Human Resources (Developers, Testers, Project Managers, Analysts)

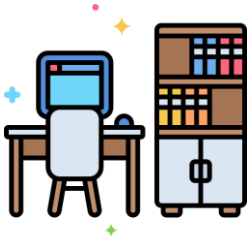




Software Tools and Cloud Infrastructure



External Contractors and Consultants



Physical Resources (Servers, Laptops, Networking Devices, Office Facilities)



Acquiring Resources(Contd.)

Key Activities in Acquiring Resources

- **Negotiating for Resources**
  - Coordinate with functional managers, vendors, or external suppliers to obtain required resources.
- **Assigning Team Members**
  - Allocate individuals based on skills, experience, and availability.
- **Obtaining Physical Resources**
  - Acquire equipment, software, and facilities required for project execution.
- **Managing Resource Availability**
  - Ensure resources are available according to project timelines and priorities.
- **Onboarding and Orientation**
  - Introduce new team members to project objectives, processes, and expectations.

Main Activities in Resource Management



Developing the Team

- Developing the Team is the process of improving the competencies, skills, collaboration, and overall effectiveness of project team members to enhance project performance and achieve project objectives successfully.

- Team development focuses on creating a motivated, collaborative, and high-performing project environment that supports value delivery.



Developing the Team (Contd.)

Key Activities in Developing the Team

- **Providing Training and Skill Development**
  - Organize technical training, workshops, and mentoring programs.
- **Conducting Team-Building Activities**
  - Encourage collaboration, trust, and positive working relationships.
- **Improving Communication**
  - Establish effective communication channels and feedback mechanisms.
- **Recognizing and Rewarding Performance**
  - Motivate team members through recognition and performance appreciation.
- **Managing Team Dynamics**
  - Address conflicts, improve cooperation, and support team cohesion.

Main Activities in Resource Management



Managing the Team

- Managing the Team is the process of monitoring team member performance, providing feedback, resolving conflicts, and supporting collaboration to improve overall project performance.

- Effective team management helps create a productive, motivated, and collaborative work environment that supports successful project delivery and stakeholder value.



Managing the Team (Contd.)

Key Activities in Managing the Team

- **Monitoring Team Performance**
  - Evaluate individual and team progress against project objectives.
- **Providing Feedback and Support**
  - Conduct performance discussions, coaching, and mentoring.
- **Resolving Conflicts**
  - Address disagreements and communication issues within the team.
- **Motivating Team Members**
  - Encourage engagement, recognition, and continuous improvement.
- **Facilitating Collaboration**
  - Promote teamwork, knowledge sharing, and problem-solving.

Main Activities in Resource Management



Controlling Resources

- Controlling Resources is the process of monitoring and managing the use of project resources to ensure that they are available, allocated properly, and utilized efficiently according to the project plan.

- This process helps optimize resource usage, reduce waste, and ensure that project activities are completed effectively and on schedule.



Controlling Resources (Contd.)

Key Activities in Controlling Resources

- **Monitoring Resource Utilization**
  - Track how human and physical resources are being used throughout the project.
- **Comparing Actual vs Planned Resource Usage**
  - Analyze differences between planned and actual resource allocation.
- **Managing Resource Conflicts**
  - Resolve issues such as resource shortages, overlaps, or scheduling conflicts.
- **Optimizing Resource Allocation**
  - Reassign or adjust resources to improve efficiency and productivity.
- **Maintaining Resource Records**
  - Update resource documentation, schedules, and performance information.



In Class Activity 2

Multiple Choice Question

Which Resource Management process focuses on improving team competencies, collaboration, and overall team performance?

A. Acquire Resources

B. Develop Team

C. Control Resources

D. Estimate Activity Resources



Multiple Choice Question

A project manager notices that several software engineers are assigned to multiple projects simultaneously, causing workload conflicts and reduced productivity. Which Resource Management process is MOST directly responsible for resolving this issue?

A. Plan Resource Management

B. Develop Team

C. Control Resources

D. Manage Team



Multiple Choice Question

In a cloud platform development project at WSO2, the project manager uses Agile sprint planning, skill matrices, and capacity planning to allocate work efficiently among developers. Which Resource Management concept is BEST represented in this scenario?

A. Procurement Management

B. Resource Optimization

C. Scope Validation

D. Risk Transfer



What is Schedule Management?

- Schedule Management in project management refers to the processes involved in planning, developing, monitoring, and controlling the project timeline to ensure that project activities are completed on time.
- Schedule management helps organizations deliver project outcomes efficiently by coordinating activities, resources, dependencies, and deadlines throughout the project lifecycle.



Common Scheduling Techniques

- **Critical Path Method (CPM)**: A scheduling technique used to identify the longest sequence of dependent activities that determines the minimum project completion time.
- **Gantt Charts**: Visual project scheduling charts that display project activities, durations, timelines, and progress using horizontal bars.
- **Network Diagrams**: Graphical representations showing the sequence, dependencies, and relationships among project activities.
- **Agile Sprint Planning**: A planning process in Agile projects where teams select and organize tasks to be completed within a short development cycle called a sprint.
- **Rolling Wave Planning**: A progressive planning technique where near-term work is planned in detail while future activities are planned at a higher level until more information becomes available.
- **Earned Value Management (EVM)**: A project performance measurement technique that integrates scope, schedule, and cost data to evaluate project progress and performance.

Main Activities in Schedule Management



Planning Schedule Management



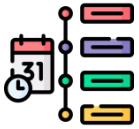
Defining Activities



Sequencing Activities



Estimating Activity Duration



Developing the Schedule



Controlling the Schedule

Main Activities in Schedule Management



Planning Schedule Management

- Planning Schedule Management is the process of establishing the policies, procedures, tools, and methodologies used to plan, develop, manage, execute, and control the project schedule.

- provides a structured approach for managing project timelines effectively and ensuring timely project delivery.



Planning Schedule Management (Contd.)

Key Activities in Planning Schedule Management

- **Defining Scheduling Approach**
  - Select scheduling methodologies such as Predictive, Agile, or Hybrid approaches.
- **Establishing Scheduling Procedures**
  - Define rules for schedule updates, approvals, and reporting.
- **Selecting Scheduling Tools**
  - Identify tools such as Microsoft Project, Jira, or Primavera.
- **Defining Performance Measurement Methods**
  - Establish schedule performance metrics and reporting mechanisms.
- **Setting Schedule Baselines**
  - Define approved timelines and milestones for monitoring project progress.

Main Activities in Schedule Management



Defining Activities

- Defining Activities is the process of identifying and documenting the specific tasks that must be performed to produce project deliverables and achieve project objectives.

- This process helps break down project work into manageable activities that can be scheduled, monitored, and controlled effectively.

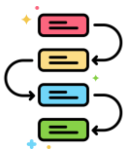


Defining Activities (Contd.)

Key Activities in Defining Activities

- **Reviewing the Work Breakdown Structure (WBS)**
  - Analyze work packages and deliverables.
- **Breaking Down Work Packages**
  - Divide large work components into detailed activities.
- **Identifying Milestones**
  - Define significant project events and checkpoints.
- **Documenting Activity Details**
  - Record activity descriptions, dependencies, and constraints.
- **Preparing Activity Lists**
  - Create a complete list of project activities for scheduling purposes.

Main Activities in Schedule Management



Sequencing Activities

- Sequencing Activities is the process of identifying and documenting the logical relationships and dependencies among project activities.

- This process helps determine the correct order of work to support efficient scheduling and successful project completion.



Sequencing Activities (Contd.)

Key Activities in Sequencing Activities

- **Identifying Activity Dependencies**
    - Determine relationships between project activities.
- **Determining Activity Order**
    - Arrange activities in the correct execution sequence.
- **Analyzing Constraints and Assumptions**
    - Consider technical, organizational, and resource-related limitations.
- **Preparing Network Diagrams**
    - Visualize activity relationships and workflow sequences.



Sequencing Activities (Contd.)

Common Dependency Types in Project Scheduling

- **Finish-to-Start (FS)**
  - Finish-to-Start is the most common dependency type where a successor activity cannot start until the predecessor activity has finished.
  - **Example:** In a software project, **system testing** cannot begin until **software development** is completed.
  - **Representation:** Activity A → Activity B (B starts after A finishes)

- **Start-to-Start (SS)**
  - Start-to-Start is a dependency where a successor activity cannot start until the predecessor activity has started.
  - **Example:** In a mobile application project, **coding** and **database configuration** may start at the same time after project setup begins.
  - **Representation:** Activity A ⇢ Activity B (B starts after A starts)



Sequencing Activities (Contd.)

Common Dependency Types in Project Scheduling

- **Finish-to-Finish (FF)**
  - Finish-to-Finish is a dependency where a successor activity cannot finish until the predecessor activity has finished.
  - **Example:** In a software release project, **documentation updates** cannot be completed until **system testing** is finished.
  - **Representation:** Activity A ⇢ Activity B (B finishes after A finishes)

- **Start-to-Finish (SF)**
  - Start-to-Finish is a rare dependency where a successor activity cannot finish until the predecessor activity has started.
  - **Example:** In an IT support transition project, the **old support system** cannot stop operations until the **new support system** has started functioning.
  - **Representation:** Activity A ⇢ Activity B (B finishes after A starts)

Main Activities in Schedule Management



Estimating Activity Duration

- Estimating Activity Durations is the process of predicting the amount of time required to complete individual project activities using available resources and project information.

- This process helps project managers develop realistic schedules and improve project planning accuracy.



Estimate Activity Durations(Contd.)

Key Activities in Estimating Activity Durations

- **Reviewing Project Activities**
    - Analyze the scope and complexity of project tasks.
- **Assessing Resource Availability**
    - Consider the skills, quantity, and availability of resources.
- **Applying Estimation Techniques**
    - Use suitable methods to estimate activity durations.
- **Considering Risks and Constraints**
    - Account for uncertainties, dependencies, and external factors.
- **Documenting Duration Estimates**
    - Record estimated durations and supporting assumptions.



Estimate Activity Durations (Contd.)

Duration Estimate Techniques

- **Analogous Estimating**
  - A technique that uses historical data from similar past projects to estimate activity durations or costs. It is quick and useful in early project stages but may be less accurate if projects are not closely similar.
  - **Example:** If a previous mobile app project required 4 weeks for UI development, a similar duration may be estimated for a new project.

- **Parametric Estimating**
  - A technique that uses statistical relationships and measurable variables to estimate durations or costs. It is more accurate when reliable historical data is available.
  - **Example:** If a developer writes 100 lines of code per day and a module requires 1,000 lines, the estimated duration is 10 days.



Estimate Activity Durations (Contd.)

Duration Estimate Techniques

- **Three-Point Estimating**
  - A technique that uses Optimistic (O), Most Likely (M), and Pessimistic (P) estimates to calculate a more realistic activity duration while considering uncertainty and risk.

$$\boldsymbol{E = \frac{O + 4M + P}{6}}$$

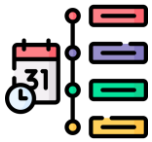
  - **Example:** For a testing activity - Optimistic = 2 days, Most Likely = 4 days, Pessimistic = 8 days

$$\text{Estimated duration } (E) = \frac{2 + 4(4) + 8}{6} = 4.33 \textit{ days}$$

- **Bottom-Up Estimating**
  - A detailed estimation technique where individual activities are estimated separately and combined to calculate the total project duration or cost. It is highly accurate but time-consuming.
  - **Example:** UI Design = 3 days, Coding = 5 days, Testing = 2 days

    Total estimated duration = 10 days.

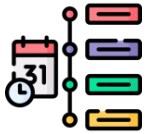
Main Activities in Schedule Management



Develop Schedule

- Develop Schedule is the process of analyzing activity sequences, durations, resource requirements, and scheduling constraints to create the project schedule model and overall project timeline.

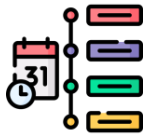
- This process helps establish a realistic and achievable schedule for completing project activities and delivering project outcomes on time.



Develop Schedule (Contd.)

Key Activities in Develop Schedule

- **Analyzing Activity Sequences**
  - Review dependencies and logical relationships among activities.
- **Reviewing Activity Durations**
  - Use estimated durations to determine the overall project timeline.
- **Allocating Resources**
  - Ensure resources are assigned according to availability and workload.
- **Applying Scheduling Techniques**
  - Use scheduling methods and tools to optimize the project timeline.
- **Establishing the Schedule Baseline**
  - Finalize and approve the project schedule for monitoring progress.

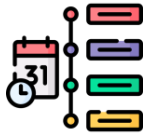


Develop Schedule (Contd.)

Critical Path Method (CPM)

- The Critical Path Method (CPM) is a schedule analysis technique used to identify the longest sequence of dependent activities that determines the shortest possible project duration.

- It helps project managers identify critical activities that must be closely managed to avoid project delays.



Develop Schedule (Contd.)

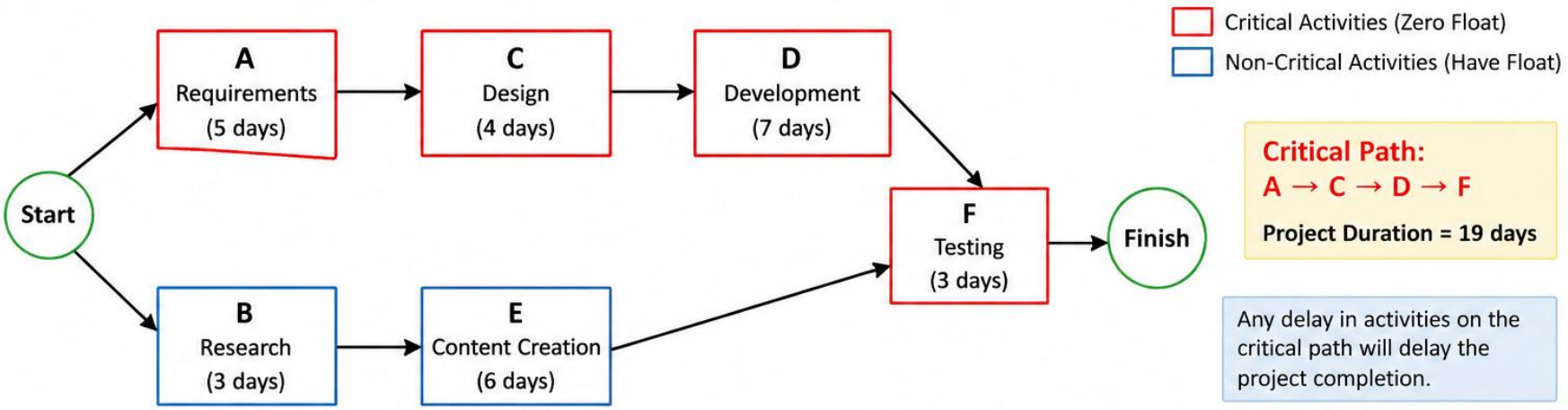
Critical Path Method (CPM)

- **Key Concepts in CPM**
  - **Critical Path** – Sequence of activities that defines total project duration
  - **Float (Slack)** – Time an activity can be delayed without affecting the project
  - **ES/EF & LS/LF** – Early and late start/finish times used for schedule calculation
- **Steps in CPM**
  1. List project activities
  2. Define dependencies
  3. Estimate durations
  4. Create network diagram
  5. Calculate ES, EF, LS, LF
  6. Identify the critical path

Develop Schedule (Contd.)

Critical Path Method (CPM) – Example

Project: Website Development



Activity	Duration (days)	Early Times		Late Times		Float (LS – ES)
		ES	EF	LS	LF	
A (Requirements)	5	0	5	0	5	0
B (Research)	3	0	3	2	5	2
C (Design)	4	5	9	5	9	0
D (Development)	7	9	16	9	16	0
E (Content Creation)	6	3	9	10	16	7
F (Testing)	3	16	19	16	19	0

Note: The critical path is A – C – D – F with a total duration of 19 days.

Main Activities in Schedule Management



Controlling the Schedule

- Controlling the Schedule is the process of monitoring project schedule performance, managing schedule changes, and taking corrective actions to ensure the project is completed on time.

- This process helps project managers compare actual progress with the planned schedule and respond effectively to delays or schedule variances.



Controlling the Schedule (Contd.)

Key Activities in Controlling the Schedule

- **Tracking Project Progress**
  - Measure actual work completed against planned activities.
- **Analyzing Schedule Variances**
  - Identify differences between planned and actual performance.
- **Managing Schedule Changes**
  - Review and implement approved schedule modifications.
- **Applying Corrective Actions**
  - Adjust resources, timelines, or activities to reduce delays.
- **Updating Schedule Information**
  - Revise schedules, forecasts, and performance reports.



Controlling the Schedule (Contd.)

Earned Value Management (EVM) in Monitoring the Schedule Performance

- Earned Value Management (EVM) is a project performance measurement technique that integrates scope, schedule, and cost to evaluate project progress and performance. It supports project monitoring, forecasting, and decision-making.

Key Components

- **Planned Value (PV)** – Budgeted value of planned work
- **Earned Value (EV)** – Value of completed work
- **Actual Cost (AC)** – Actual cost spent on completed work



Controlling the Schedule (Contd.)

Important EVM Metrics

- **Schedule Variance (SV):** Measures schedule performance.

$$SV = EV - PV$$

  - **SV = 0** → On schedule, **SV > 0** → Ahead of schedule, **SV < 0** → Behind schedule

- **Schedule Performance Index (SPI):** Measures schedule efficiency.

$$SPI = \frac{EV}{PV}$$

  - **SPI = 1** → On schedule, **SPI > 1** → Ahead of schedule, **SPI < 1** → Behind schedule

- **Cost Variance (CV):** Measures cost performance.

$$CV = EV - AC$$

  - **CV > 0** → Under budget, **CV < 0** → Over budget

- **Cost Performance Index (CPI):** Measures cost efficiency.

$$CPI = \frac{EV}{AC}$$

  - **CPI > 1** → Cost efficient, **CPI < 1** → Cost overrun



Controlling the Schedule (Contd.)

EVM Example

- If:
  - PV = $100,000
  - EV = $90,000
  - AC = $110,000
- Then:
  - $SV = EV - PV = \$90{,}000 - \$100{,}000 = \mathbf{-10{,}000} \rightarrow \mathbf{SV < 0}$
  - $CV = EV - AC = \$90{,}000 - \$110{,}000 = \mathbf{-20{,}000} \rightarrow \mathbf{CV < 0}$
  - $SPI = \frac{EV}{PV} = \frac{\$90{,}000}{\$100{,}000} = \mathbf{0.9} \rightarrow \mathbf{SPI < 1}$
  - $CPI = \frac{EV}{AC} = \frac{\$90{,}000}{\$110{,}000} = \mathbf{0.82} \rightarrow \mathbf{CPI < 1}$

This indicates the project is behind schedule and over budget.



In Class Activity 3

Multiple Choice Question

Which scheduling technique identifies the longest sequence of dependent activities that determines the minimum project completion time?

A. Rolling Wave Planning

B. Earned Value Management (EVM)

C. Critical Path Method (CPM)

D. Agile Sprint Planning



Multiple Choice Question

A project has the following EVM values:

- **Earned Value (EV) = $80,000**
- **Planned Value (PV) = $100,000**

What does the Schedule Performance Index (SPI) indicate?

A. The project is ahead of schedule

B. The project is on schedule

C. The project is behind schedule

D. The project is under budget



Multiple Choice Question

Which dependency type indicates that an activity cannot start until its predecessor activity has finished?

A. Start-to-Start (SS)

B. Finish-to-Finish (FF)

C. Start-to-Finish (SF)

D. Finish-to-Start (FS)



Summary

- **Procurement Management** focuses on acquiring products, services, or resources from external suppliers through processes such as planning, conducting, controlling, and closing procurements.
- **Resource Management** involves planning, acquiring, developing, managing, and controlling human and physical resources to improve project performance and productivity.
- **Schedule Management** ensures project activities are planned, sequenced, monitored, and controlled to achieve timely project completion.
- **Planning Processes** establish management approaches, procedures, tools, and guidelines for procurement, resources, and scheduling activities.
- **Activity Definition and Sequencing** help identify project tasks, dependencies, and workflow relationships necessary for effective scheduling.
- **Resource and Duration Estimation Techniques** such as Analogous Estimating, Parametric Estimating, Three-Point Estimating, and Bottom-Up Estimating improve planning accuracy.
- **Critical Path Method (CPM)** identifies the longest sequence of dependent activities that determines the minimum project completion time.
- **Earned Value Management (EVM)** integrates scope, cost, and schedule data to measure project performance using metrics such as SV, SPI, CV, and CPI.